

Autonomous Driving Project – Team Task Division

#	Task	Short Summary	Ali Abdelkhalek	Mohamed Awis	Yassin Elhalawany
1	Git & Project Setup	Create Git repo and organize project	Created the Task devision File	Created the Git repo	Organized the files for the first meeting
2	System Engineering	Develop SysML diagrams and requirements	Created Requirements diagaram and Internal Block Diagram (IBD)	Created Use case diagaram, Activity diagram and Package diagram	Created Sequence diagaram and Block Definition Diagram (BDD)
3	Prototype Simulation	Develop first Tinkercad prototype		Created TinkerCad simulation circuit (with coding for detecting the line only) and electrical circuit	
4	Line Detection	Vehicle follows lane autonomously	Performed cable management and wire routing, organized and assembled components, optimized the code to suit the available hardware conditions, implemented black line detection and tracking, calibrated IR sensors, integrated system components, Speed Optimization and Performance Tuning for Improved Line-Tracking Accuracy and Stability and conducted real-world testing and performance evaluation of the car.		
5	Milestone 1	The Vehicle Should follows lane			
6	Routing Functions	Oval and figure-eight routing			
7	Parking Function	Autonomous parking capability			
8	Obstacle Detection	Detect and avoid obstacles			
9	Milestone 2	Vehicle detects obstacles successfully			